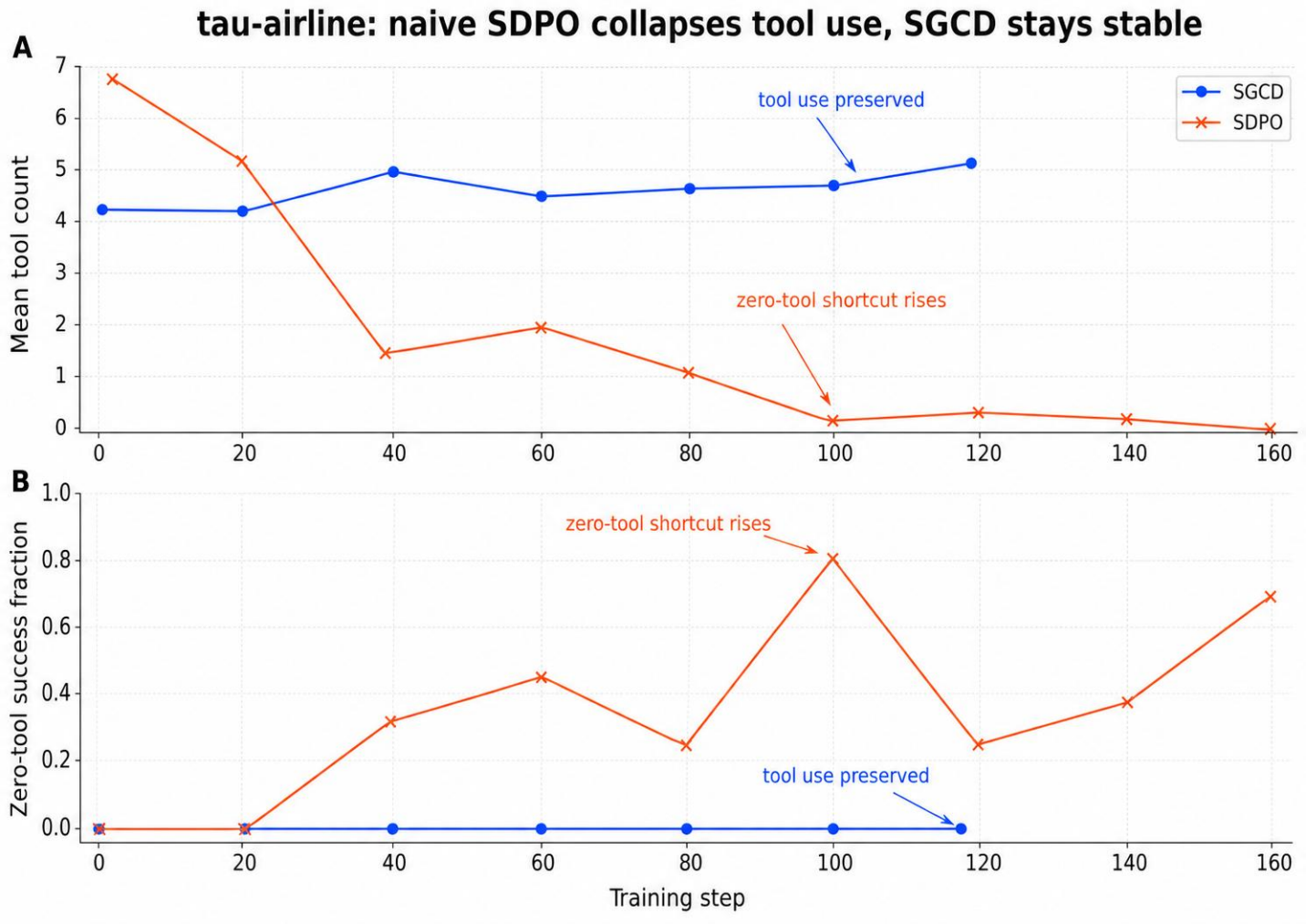

tau-airline: naive SDPO collapses tool use, SGCD stays stable
A
Mean tool count
SGCD
SDPO
tool use preserved
zero-tool shortcut rises
B
Zero-tool success fraction
zero-tool shortcut rises
tool use preserved
Training step